Online Adaptive Correction of a Frozen Vision–Language–Action Model

Theory, method, and results

Taheri et al.

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Result in one line

An actuator fault in a frozen 3.35 B vision–language–action model is identified and corrected *within a single episode*, from the robot's own motion, with no gradients through the task and no search over task success — **across four LIBERO suites, 29% → 73% pooled, $p = 1.1 \times 10^{-7}$ — provided the correction is restricted to the channels on which the fault is identifiable.** Correcting the rest is worse than doing nothing.

The scope condition is the contribution. With all six action channels corrected, the method is significant on *one suite of four*. Restricted to identifiable channels it is significant on *all four*. Identifiability is predictable before deployment from healthy data alone.

Artifacts. Video: `results/phase05/adaptive_vs_frozen.mp4`. Data: `results/`. Code: `openpi/{error_signal, openloop_id, adaptive_law, compare_video}.py`. Full theory: `docs/ADAPTIVE_CONTROL_VLA.md`.
Status. Simulation; hardware validation in progress. Section 13 states what this does not establish.

Contents

1 The problem

A vision–language–action (VLA) policy deployed on a robot degrades when the hardware does: an actuator develops a constant offset, a joint loses gain, a sensor drifts. The policy was trained on healthy hardware and systematically commands the wrong thing.

Retraining a 3.35 B-parameter policy is not possible mid-deployment, and often not possible at all — the operator typically owns neither the weights nor the training data. Black-box search over task success avoids retraining but discards everything known about the plant and pays for it in episodes.

The observation this rests on: *the robot measures its own end-effector pose every control step.* If the fault is observable there it can be identified directly, inside one episode, without consulting the task reward.

The finding that organises everything below is that "observable there" is **channel-dependent**, decidable in advance, and that acting on channels which fail the test is actively harmful.

2 Setting

- **Model.** $\pi_{0.5}$-LIBERO, frozen: 3.35 B parameters, no fine-tuning, no backpropagation through the environment. Faults are injected client-side between the policy and `env.step`; the model is never modified.
- **Benchmarks.** `libero_spatial` (nominal **99.0%** over 500 episodes), plus `libero_goal`, `libero_object`, `libero_10`.

- **Controller.** OSC_POSE at 20 Hz, `control_delta = True`, `output_max` = $[0.05, 0.05, 0.05, 0.5, 0.5, 0.5]$ (m, m, m, rad, rad, rad).
- **Controls.** Frozen and adaptive arms share task, initial state, policy and fault; the only difference is whether the law runs.

3 Fault screening

Nine conditions screened (20 episodes each) for one that hurts enough to matter and leaves the robot recoverable.

Fault	**Severity**	**Success**	**Verdict**
nominal	—	95 %	reference
brightness	0.1 / 0.2 / 0.4	100 %	no effect at any level
gain	0.7	95 %	no effect
gain	0.5	75 %	mild
gain	0.3	0 %	catastrophic
offset	**0.05**	**55 %**	**degraded, recoverable**
offset	0.10	5 %	near-lethal
offset	0.20	0 %	catastrophic

Visual perturbation does nothing at any level tested; actuator faults show a sharp cliff. Offset 0.05 — a constant $f = +0.05$ added to the six arm dimensions of every action, i.e. a miscalibrated actuator — is the primary condition throughout.

4 Is the fault class reachable? The oracle

Before attempting online adaptation we established that the fault is repairable in principle by an edit at one site. A six-vector computed analytically was added to `action_out_proj/bias`, scaled by k (15 episodes per condition).

Condition	**Success**	**vs. floor**
$k = -1.0$ (wrong sign), faulted	0.0 % (0/15)	−46.7
$k = 0.0$ (no edit), faulted *(floor)*	46.7 % (7/15)	—
$k = 0.5$, faulted	86.7 % (13/15)	+40.0
$k = 1.0$ **(computed oracle), faulted**	**100.0 % (15/15)**	**+53.3**
$k = 1.5$, faulted	20.0 % (3/15)	−26.7
$k = 1.0$ on the *healthy* policy	6.7 % (1/15)	—

Complete.
46.7% → 100% against a 99.0% nominal — the entire deficit, from a fixed six-vector on one bias. No gradient, no fine-tuning.

Specific, not a general improvement.
The same edit on the *healthy* policy collapses it from 99% to 6.7%; sign-flipped it annihilates the faulted one (0%). A compensating inverse, not a tonic.

Narrow basin.
$k = 1.5$ scores 20% — *worse than not repairing.* Any search over this class must be scaled to $k \in [0.4, 1.2]$.

Why the edit must be computed per dimension. "Cancel a +0.05 offset with a −0.05 edit" is wrong, and would have read as the class being unreachable. $\pi_{0.5}$ emits *normalised* actions; `Unnormalize` runs afterwards, so with env $= \frac{\text{norm}+1}{2}(q_{99} - q_{01}) + q_{01}$ each channel carries its own scale, and those scales differ by **7**×. A uniform fault in environment space is wildly **anisotropic** in the model's units. This same anisotropy turns out to decide the whole scope condition (§7).

5 Theory of the online law

5.1 The three objects

An adaptive law needs an **error** to drive to zero, a **plant model** predicting error-free behaviour, and the **map** from the unknown parameter to that error. Get any wrong and the law diverges. All three failure modes were observed here before being fixed (§11).

5.2 Plant

One action unit *commands* 0.05 m of translation. **It does not achieve it**: regressing achieved on commanded displacement over fault-free rollouts gives a static gain of **0.21–0.25**. The arm covers about a fifth of the commanded delta inside one 50 ms period, so using `output_max` as the plant is wrong by 5× and an adaptive law built on it mistakes ordinary controller lag for a fault. An FIR model over the last $K = 6$ commands captures it:

$$y_t \;=\; \sum_{k=0}^{K} h_k \, a_{t-k} \;+\; c. \tag{1}$$

R^2	dx	dy	dz	dr_x	dr_y	dr_z
static ($K = 0$)	0.885	0.961	0.978	0.416	0.091	0.077
FIR ($K = 6$)	0.975	0.982	0.982	0.489	0.109	0.168
FIR, after SO(3) fix	**0.975**	**0.982**	**0.982**	0.522	0.336	**0.972**

5.3 Error signal

With $u_t = a_t + c_t + f$ the executed action (policy command, our correction, unknown fault) and $\hat{P}$ the identified fault-free plant,

$$r_t \;=\; y_t - \hat{P}(a_t + c_t) \;\approx\; Mf. \tag{2}$$

r_t is independent of c_t *when* $\hat{P} = P$ *exactly*, so the estimator does not chase its own correction. §6 shows what happens when that condition fails, and corrects this claim.

5.4 The map

$M = \partial(\text{achieved motion})/\partial f$, measured **open loop**: a recorded command sequence replayed with and without a fault, so commands are identical by construction and any motion difference is the fault propagating. One axis at a time, ±0.02 central difference:

	fault applied to					
observed	dx	dy	dz	dr_x	dr_y	dr_z
dx	**0.297**	0.021	−0.112	−0.029	0.081	−0.006
dy	0.008	**0.272**	0.023	−0.034	0.016	0.003
dz	0.029	0.032	**0.126**	−0.037	0.012	−0.003
dr_x	−0.003	−0.001	−0.001	**0.253**	−0.004	0.002
dr_y	0.013	0.001	−0.001	−0.007	**0.276**	−0.000
dr_z	0.001	0.004	−0.003	0.011	0.005	**0.244**

Diagonal 0.13–0.30 (the plant damps the fault 3–8×, so the law needs ≈ 4× gain); off-diagonal mass 26% of $|M|$, dominated by a real $dx \leftrightarrow dz$ term; $\mathrm{cond}(M) = 3.0$.

5.5 Orientation increments: a metric bug that corrupted four results

Rotation change must be the **relative rotation**, not a difference of axis–angle vectors: $\omega = \mathrm{axis\_angle}(q_{t+1} \otimes \mathrm{conj}(q_t))$. Axis–angle is a *chart*, not a vector space: the same rotation has representations differing by 2π and the chart is singular at 0 and π. Over 2000 random 0.05 rad increments the relative rotation recovers truth to 1.4×10^{-14}; the difference of charts errs by up to **6.28** — a full wraparound.

Retraction (2026-08-27)

An earlier version reported the rotation block of M as “a swapped, sign-flipped pair” and argued a per-axis law would therefore diverge. **That was wrong** — an artifact of the chart difference. With the correct metric the rotation axes have ordinary diagonal gains of 0.276 and 0.244, off-diagonal mass falls from 0.59 to 0.26, and the divergence argument does not apply. Withdrawn.

The wrong metric corrupted **four separate results**: rotation appeared unpredictable, M appeared cross-coupled, the gain law's rotation estimates were meaningless, and deliberate excitation made performance *worse*. One bug.

5.6 The law

$$\hat{f} \leftarrow \mathrm{clip}\left(\hat{f} + \gamma\left(\frac{M^{-1} r_t}{1 + \|r_t\|^2/\rho^2} - \hat{f}\right), \pm 0.15\right), \qquad c_t = -\hat{f} \tag{3}$$

with a deadzone for $\|r_t\| < \delta$; $\gamma = 0.08$, $\rho = 0.15$, $\delta = 0.008$. Each robustness term earned its place by fixing an observed failure (§11), not by assumption.

6 What the correction is actually doing

6.1 $\hat{f}$ alone cannot distinguish identification from bias

Run the law with **no fault at all** and $\hat{f}$ does not go to zero. It settles at $[+0.045, +0.014, +0.073, -0.009, +0.017$ — on dz that is 146% of the magnitude of the real fault. The estimator reports a large offset where there is nothing to correct.

The diagnostic that separates them is the **separation**: the estimate under a fault minus the estimate with no fault, on matched episodes.

	no fault	fault 0.05	separation	true
dx	+0.045	+0.048	+**0.003**	0.050
dy	+0.014	+0.038	+0.023	0.050
dz	+0.073	+0.079	+**0.006**	0.050
$\mathbf{dr_x}$	−0.009	+0.041	+**0.049**	0.050
dr_y	+0.017	+0.033	+0.016	0.050
$\mathbf{dr_z}$	−0.001	+0.047	+**0.048**	0.050

Rotation identifies the fault almost exactly. Translation does not identify it at all.

Method note. No estimate of a fault parameter should be believed without a matched no-fault control. $\hat{f}$ looked convincing on all six axes for weeks of runs; only the separation revealed that half of it was bias.

6.2 Where the phantom comes from

Measured separately with `--estimate-only` (update $\hat{f}$, never apply it): mean |phantom| is **0.0143** open loop and **0.0266** closed loop.

Half is plant-model error, and it is not task-specific — identifying the plant on 10 tasks instead of 3 made the phantom *worse* (0.027 → 0.034). *Half is estimator feedback, which corrects the claim in §5.3*: r_t is independent of c_t only when $\hat{P} = P$ exactly. With model error $r = Mf + \varepsilon(a + c)$, so $\hat{f}$ converges to $f + M^{-1}\varepsilon(a + c)$ — it does chase its own correction. Measured amplification: **1.9**×.

7 The scope condition: correct only what you can identify

7.1 With all six channels corrected, the method works on one suite of four

Suite	frozen	corrected	Δ	p
`libero_spatial`	18/40 = 45%	38/40 = 95%	+50	< **0.0001**
`libero_goal`	8/20 = 40%	12/20 = 60%	+20	0.34
`libero_object`	7/20 = 35%	9/20 = 45%	+10	0.75
`libero_10`	0/20 = 0%	3/20 = 15%	+15	0.23
pooled, 3 new suites	15/60 = 25%	24/60 = 40%	+15	0.12

It is **not** an identification failure: plant fits on the new suites are comparable or better (`object` reaches rotation R^2 0.771/0.692 against spatial's 0.522/0.336). The estimator still works; the *task* does not recover.

7.2 Restricted to identifiable channels, it works on all four

Suite	frozen	corrected	Δ	p
`libero_spatial`	7/15 = 47%	14/15 = 93%	+47	**0.014**
`libero_goal`	9/20 = 45%	18/20 = 90%	+45	**0.006**
`libero_object`	6/20 = 30%	16/20 = 80%	+50	**0.004**
`libero_10`	0/20 = 0%	7/20 = 35%	+35	**0.008**
pooled	**22/75 = 29%**	**55/75 = 73%**	+44	$\mathbf{1.1 \times 10^{-7}}$

`libero_10` is the sharpest case: a long-horizon suite (520-step cap) where the fault takes the policy to **zero**, and correcting three channels recovers 7/20 from that floor.

7.3 Non-identifiable channels do not merely fail — they do harm

On `libero_object`, three arms on the same fault and the same episodes:

Correction	frozen	corrected	Δ
all six dims	7/20	9/20	+10
rotation only	6/20	**16/20**	**+50**
translation only	6/20	**6/20**	**0**

Translation alone contributes **exactly nothing**, and including it drags a +50 effect down to +10. On `object` the translation estimates are **sign-wrong** ($-0.031, -0.018$ against $+0.050$), so that correction pushes the arm the wrong way. On `spatial` the same wrong correction happened to be harmless — which is why the naive version looked suite-specific rather than simply misapplied.

7.4 Identifiability is predictable before deployment

Nothing here requires knowing the fault. Three quantities, all measurable on *healthy* data, decide which channels are identifiable:

1. **The action normalisation** — quantile scales from the checkpoint. A uniform env-space offset is 3% of the action range on translation and **19.5%** on rotation, 6.6× larger.
2. **The plant fit** — per-channel FIR R^2 on fault-free rollouts. Identification quality tracks it in every experiment run.
3. **The policy's command statistics** — an additive fault is loud where the quantile scale is small; a multiplicative one is loud where the command is large.

The rule. Measure identifiability on healthy data, correct only those channels, leave the rest alone. The all-channel table is the ablation showing what it costs to ignore this.

8 Identifiability, stated properly

8.1 Proposition 1 — a constant input fault and a constant output bias are not separable

Suppose the measurement carries an unknown constant bias b: $y_t = M(a_t + f) + b + \text{noise}$. Within one episode f and b are both constant, so they enter only through the sum $Mf + b$. M is square and full rank, so for *any* b' the alternative fault $f' = f + M^{-1}(b - b')$ reproduces every observation exactly. The pair (f, b) is unidentifiable; only $Mf + b$ is. **The bias must come from outside the episode** — it is not a nuisance parameter one can fit.

Corollary (onset breaks the degeneracy). If the fault switches on at t_0, with $f = 0$ for $t < t_0$, then b is identified on the pre-onset window and f on the post-onset window. **Mid-episode onset is not a robustness flourish — it is what makes the problem well-posed**, and it is the honest deployment story: the estimator needs to have seen the healthy plant.

8.2 Proposition 2 — the two fault classes are loud on complementary channels

With quantile normalisation and $R_i = (q_{99} - q_{01})_i$: an **additive** env-space offset δ appears in normalised units as $2\delta/R_i$, so its signal *falls* with R_i. A **multiplicative** fault has information matrix $\sum_t \phi^\top \phi = \mathrm{diag}(\sum_t a_{t,i}^2)$, so its signal *rises* with command magnitude, and command spread is what R_i measures.

ch	$q_{99} - q_{01}$	additive signal	multiplicative signal
x	1.685	0.059	0.899
y	1.656	0.060	0.883
z	1.875	0.053	1.000
r_x	0.256	**0.391**	0.137
r_y	0.351	**0.285**	0.187
r_z	0.506	**0.198**	0.270
translation mean		0.058	**0.927**

The same quantity that makes a channel quiet for an offset makes it loud for a gain. This is a prediction, not a fit: it says an offset fault should be correctable on rotation and a gain fault on translation, which is what §7 and §10 find.

9 Results

9.1 Headline, confirmatory

Arm	Success	95% CI
nominal (no fault)	99 %	—
frozen, faulted	**18/40 = 45 %**	[31, 60]
adaptive	**38/40 = 95 %**	[83, 99]

+50 points, Fisher $p = 1.1 \times 10^{-6}$, on initial states 40–43 that no earlier run touched. The earlier n=15 estimate (47% → 93%, $p = 0.014$) did not shrink under retest — it grew slightly, and the estimates reproduced within ±0.004 on all six axes across two independent runs.

9.2 The full condition set

Condition	frozen	adaptive	note
offset 0.05, from step 1 (n=40)	45%	**95%**	$p = 1.1\text{e-}6$
offset 0.10, near-lethal (n=8)	0%	38%	frozen fails every episode
structured $[0, 0, 0, +.06, -.06, +.03]$ (n=12)	17%	**83%**	shape never tuned on
onset at step 15 (n=20)	60%	**90%**	$p = 0.065$; tracking
onset at step 45 (n=15)	73%	80%	little headroom by design
no fault (n=15)	100%	**100%**	safe to leave running
gain 0.5, multiplicative (n=10)	50–70%	70–90%	noise-dominated

9.3 Generalisation to an untuned fault shape

The structured fault $[0, 0, 0, +0.06, -0.06, +0.03]$ — mixed signs, rotation only, unseen magnitudes — is recovered as a *pattern*, not a scalar:

dim	true	separation	error
dr_x	+0.060	+0.053	−0.007
dr_y	−0.060	−0.038	+0.022
dr_z	+0.030	+0.029	−0.001
dx, dy, dz	0.000	−0.003, −0.024, −0.006	≈ 0

Sign correct on 3/3 including the negative axis, and correctly ≈ 0 where the fault is zero. This rules out the objection that the method was fitted to one fault shape.

9.4 Tracking a fault that appears mid-episode

The deployment case — a healthy robot that breaks while running. Fault injected at control step 15; mean $\hat{f}$ on identifiable rotation channels over 20 episodes:

window	$\hat{f}$
steps 0–14, before onset	**+0.009**
steps 15–30, just after	+0.035
steps 30–50, settling	+0.041
steps 50–78, late	**+0.042** (true 0.050)

Quiet before the fault exists, **70% of truth within 15 control steps (0.75 s)**, settling at 84% and holding flat with no post-convergence drift. This is also the cleanest control in the set: **the episode is its own baseline**, so the estimator bias of §6 cancels without a separate matched run.

10 The gain fault: a state-dependent fault needs a regressor

A constant offset is the easy case — constant, so a constant correction cancels it. A gain error $u = g\,a$ is state-dependent and needs a regressor: with $\beta = g - 1$, $z = M^{-1}r \approx \beta \odot \psi$, $\psi = a + c$.

Gain fault $g = 0.5$ (true $\beta = -0.50$), 10 episodes:

Variant	Adaptive	Frozen	transl. β	rot. β	rot upd/ep
LMS, old metric	7/10	6/10	−0.589	+0.016	2
RLS + dither, old metric	5/10	7/10	−0.617	+0.051	136
RLS + dither, SO(3) fix	**7/10**	5/10	−0.636	−0.169	30

Per-dimension β after the fix: dx −0.62, dy −0.49, dz −0.80, dr_x −0.20, dr_y +0.08, dr_z −0.38. The metric fix does for the gain law what it did for the offset law: dr_z flips from +0.363 — confidently the *wrong sign* — to −0.384; dy lands at −0.49 against −0.50. Four of six carry the right sign and a sensible magnitude.

Honest reading

Frozen scored 5, 6, and 7 of 10 across these three runs. **None of the adaptive-vs-frozen success differences at n=10 are resolvable.** The β columns carry the signal; the success columns are noise at this sample size.

Consistent with Proposition 2, the gain fault is identified best on *translation* — the complement of the offset fault's rotation channels.

11 Negative results worth keeping

Naive error signal, no plant model. Using $\Delta x/\texttt{output\_max} - a$ gives a nominal error of -0.175 on dx when it should be 0. Differencing faulted against nominal recovered ratios of 0.39, 2.46, -2.85 — noise. *Controller lag reads as a fault unless the plant is modelled.*

Robustness constants set without measuring. The bare law drifted — dz reached 0.642, **13× truth**. The first fix made it *worse*: a deadzone of 0.05 against a typical residual norm of 0.034 suppressed nearly every update, ending several episodes at $\hat{f} = 0$ exactly. Rescaling to the measured residual fixed both; raising the normalisation constant ($0.05 \to 0.15$, cutting a systematic 32% attenuation to $\approx 5\%$) took the result from 70% to 87%.

A coupled MIMO plant: better fit, worse extrapolation. Wrist rotation is driven substantially by *translation* commands, which a per-axis model cannot represent. A coupled plant improves leave-one-episode-out held-out R^2 from $0.107 \to 0.615$ on dr_y. **And it made the closed loop worse**: 13/15 ($p = 0.109$) with dy estimated at -0.043 against a true $+0.050$ — a sign error the per-axis model never makes — versus 14/15 ($p = 0.014$). Forty-three parameters fitted on three nominal episodes extrapolate badly once the correction moves the executed command off that distribution, and *the adaptive law lives out of distribution by construction.* Kept as `--mimo`, default off.

Excitation is not the limiting factor for rotation. The policy essentially never commands rotation: translation command sd is 0.32–0.55 with 49–74% of steps above 0.15, while all three rotation axes sit at sd 0.02–0.055 with **0.0%** above 0.15. But excitation alone does not explain dr_y: it has the most rotation excitation and the worst fit, while dr_z has less and fits at 0.972. Coupling, not excitation, is the cause.

The recurring lesson. An improvement measured in one regime — open-loop fit, or excitation — can cost you in the regime that actually matters. The MIMO plant and the dither result have the same shape.

12 The video

`results/phase05/adaptive_vs_frozen.mp4` shows two runs side by side: the same frozen model, the same task, the same injected offset fault. The only difference is whether the online repair is active. One run is the frozen policy — the fault has shifted its actions and it fails the task. The other is the same weights with the adaptive law running: it identifies the fault from the robot's own motion within the first steps and cancels it, and the task completes normally. Nothing is retrained between the two runs.

A companion clip, `adaptive_vs_frozen_offset010.mp4`, shows the harder severity (offset 0.10), where frozen success is 0/8.

13 Limits and what this does not establish

- **Simulation.** Four LIBERO suites; hardware validation is in progress and has already produced one negative — the hardware plant is run-varying on the load-bearing joints, so the offline identification recipe will not port unchanged.
- **The law is blind to sensor bias.** $\pi_{0.5}$ ignores proprioception entirely, so a proprioceptive fault produces no change in the policy's behaviour for the estimator to detect.

- **Gain faults are not settled.** One severity, n=10, no resolvable success difference; only the β estimates carry signal.
- **Rotation remains partly unidentified.** dr_y still carries the wrong sign in the gain law and fits at $R^2 = 0.336$.
- M **is identified open loop** and assumed valid in closed loop, where the policy compensates and the arm visits different states.
- **A gap to the oracle remains.** The oracle reaches 100%, the adaptive law 95% on `spatial` and 73% pooled.
- **The site screen does not separate sites** at the sample size run; `action_out_proj/bias` was chosen structurally.

14 Summary

1. A frozen 3.35 B VLA under a miscalibrated-actuator fault drops from 99% to 45% task success.
2. The fault is **exactly repairable** by a six-number edit to one layer's bias (47% → 100%), provided the edit is computed *per dimension in the model's normalised coordinates*, where a uniform physical fault is anisotropic by 7×.
3. The repair can be **identified online from proprioception alone**, within a single episode, with no gradients and no search over task success — and it tracks a fault that appears mid-episode to 70% of truth in 0.75 s.
4. **The contribution is a scope condition.** Across four suites the method is significant on one of four when all channels are corrected, and on *all four* (29% → 73%, $p = 1.1 \times 10^{-7}$) when restricted to identifiable channels. Correcting a non-identifiable channel is *worse than not correcting at all.*
5. **Identifiability is decidable in advance** from healthy data — action normalisation, plant fit, and command statistics — and the theory predicts that additive and multiplicative faults are identifiable on *complementary* channels, which the experiments bear out.
6. Getting there required a metric bug on SO(3) that had silently corrupted four results, and a matched no-fault control that revealed half of a convincing-looking estimate was bias.